Author Disclosure: M.C. Ward, None; L. Lan, None; J.J. Beitler, Oakstone Medical Publishing, A. Employment; Community Cancer Education, Inc, D. Speakers Bureau/Honoraria; Educational Concepts Group, D. Speakers Bureau/Honoraria; ACR-Image Metrix, F. Consultant/Advisory Board; American College of Radiology, F. Consultant/Advisory Board; American Board of Radiology, F. Consultant/Advisory Board; Bayer Schering Pharma AG, F. Consultant/Advisory Board; United States Army Reserve, G. Other.

2508 Juxtapapillary Choroidal Melanoma Treated with Fractionated Stereotactic Radiotherapy: Local Control and Toxicity Outcomes

R. Al Wassia[1], A. Dal Pra[1], K. Shun[1], A. Shaban[1], C. Corriveau[2], C. Edelstein[1], R. Ruo[1], H. Patrocinio[1], G. Shenouda[1]

[1]McGill University Health Center, Montreal, QC, Canada, [2]Universite de Montreal, Montreal, QC, Canada

Purpose/Objective(s): To report our experience with a LINAC-based fractionated stereotactic radiotherapy in patients with juxtapapillary choroidal melanoma.

Materials/Methods: We performed a retrospective review of 50 consecutive patients with juxtapapillary choroidal melanoma treated with stereotactic radiotherapy at our institution between April 2003 and December 2009. All medical records were reviewed to assess local control and complications. All lesions were within 2 mm from the optic disc. All patients had orbit MRI with Gadolinium for co-registration with the planning CT scans. The GTV consisted of a composed volume including the lesion in axial, coronal, and sagittal projections. The PTV was the GTV + 3mm margin. The median dose was 60 Gy (range: 54-72 Gy) with a median of 10 fractions (range: 9-12 fractions). The eye movements were monitored using an active optical system with TV monitoring during the delivery of each treatment. For the purpose of this analysis, follow-up time was calculated from the end of treatment. The results are expressed as actuarial values using the Kaplan-Meier estimates.

Results: The median follow-up was 25 months (1-72 months). There were 31 males and 29 females with a median age of 69 years (range: 30-92 years). The actuarial local control rates at 2 and 5 years were 98% and 76%, respectively. The disease-free survival rates at 2 and 5 years were 95% and 71%, respectively. Only two patients developed systemic metastases, at 11 and 27 months after treatment. Two patients died, one from metastasis and one from cardiac disease. The actuarial 2 and 5 year enucleation-free survival rates were 97% and 84%, respectively. There was 1 case (2%) of optic neuritis. The actuarial complication rates at 2 and 5 years were: 33% and 88% for radiation-induced retinopathy, 9.3% and 46.9% for dry eye, 20% and 58% for neovascular glaucoma, 30% and 90% for complete visual loss, and 12% and 53% for cataract, respectively.

Conclusions: Stereotactic fractionated radiotherapy is a very attractive non-invasive treatment for juxtapapillary choroidal melanomas with a satisfactory local control rate and an acceptable toxicity profile.

Author Disclosure: R. Al Wassia, None; A. Dal Pra, None; K. Shun, None; A. Shaban, None; C. Corriveau, None; C. Edelstein, None; R. Ruo, None; H. Patrocinio, None; G. Shenouda, None.

2509 Implementation of Virtual Endoscopy in Head and Neck Cancer Treatment Planning

A. Chan, M. Hasan, G. Chen, J. Wolfgang

Massachusetts General Hospital, Harvard Medical School, Boston, MA

Purpose/Objective(s): The anatomy of the head and neck is complex, and visualization of this region is of utmost importance for the treatment of head and neck cancers. 2-dimensional axial CT/MRI images and fiber optic endoscopy, though with limitations, are traditionally used. The purpose of this study was to develop high-resolution virtual endoscopy navigation engine using a combination of surface- and 3-dimensional (3D) volume rendering techniques, and to investigate its use in the head and neck cancer radiation treatment planning.

Materials/Methods: In-house GPU based and the commercial Fovia high-definition surface and volume rendering engines were used to construct software tools that allowed visualization of the virtual patients. Head and neck cases that were chosen for the current study consisted of advanced nasal cavity and paranasal sinus cancers, nasopharyngeal cancers, and early and late laryngeal cancers. CT treatment planning data sets were used for the volume rendered patient anatomy. Line integrals along the direction of light propagation from the observer to the region of interest determine the endoscopic trajectory, where minimum values correspond to rays traversing patient airways. Segmented data in conjunction with planned contoured volumes and calculated dose were then be overlaid on the endoscopic visualization.

Results: With the same simulation CT data, high resolution virtual endoscopy navigation movies were generated using a combination of surface and volume rendering optimization techniques. While axial images of helical treatment planning CT demonstrate the location, size, and extent of the tumors in the studied cases, virtual endoscopy images display the mucosal surface structures and the three-dimensional images and anatomical relation of the tumors. Virtual endoscopy is particularly useful in evaluating large obstructive lesions of the nasal cavity and larynx as it allows retrograde, fly-through, and fly-around navigations through the obstructive lesions, which are not feasible with fiber optic endoscopy. In addition, the walls of the nasal cavity and paranasal sinuses can be traversed, allowing the visualization of the extent of tumor. Calculated dose is mapped to the visible surfaces in the virtual patient, allowing for exploration of the proposed irradiated volume in three dimensions.

Conclusions: Surface and volume rendered virtual endoscopy can be a useful tool for examination of both the patient anatomy, and the potential impact of the radiation delivery upon diseased and normal tissue. Used in conjunction with adaptive radiotherapy techniques that employ regular volumetric and functional imaging (CT, MRI, PET), the progress of the radiotherapy treatment may be followed continuously.

Author Disclosure: A. Chan, None; M. Hasan, None; G. Chen, None; J. Wolfgang, None.

2510 How to Easily Deliver 75 Gy in 5 weeks for Advanced Head and Neck Cancer

J. Cvek[1], J. Kubes[2], B. Otahal[1], P. Kominek[1], D. Feltl[1]

[1]FN Ostrava, Ostrava, Czech Republic, [2]FN Bulovka, Praha, Czech Republic